Practical - 14

Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log

Problem Definition : Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role.

A. Theory

1. Are dashboard values loaded dynamically from the database?

Yes. All metrics on the admin dashboard (Active Students, Total Events, Total Registrations, Admin Accounts, and Audit Log count) are calculated dynamically using SQL queries (SELECT COUNT(*) FROM ...). The "Latest Registrations" table is also fetched dynamically using an INNER JOIN across registrations, students, and events.

2. Is access restricted to admin users?

Yes. Every administrative page checks $_SESSION['role'] === 'admin' at the very top of the script. If an unauthorized user (student or guest) tries to access the page directly via URL, they are immediately blocked with header("Location: access_denied.php"); exit();.

3. Are role-based menus working?

Yes. The modular navbar.php checks the session role:

- When logged in as Admin: Displays Dashboard, Students, Events, Audit Logs, and Logout (Admin).
- When logged in as Student: Displays Student Home, View Events, Try Admin Area, and Logout (Student).
- When Guest: Displays Login only.

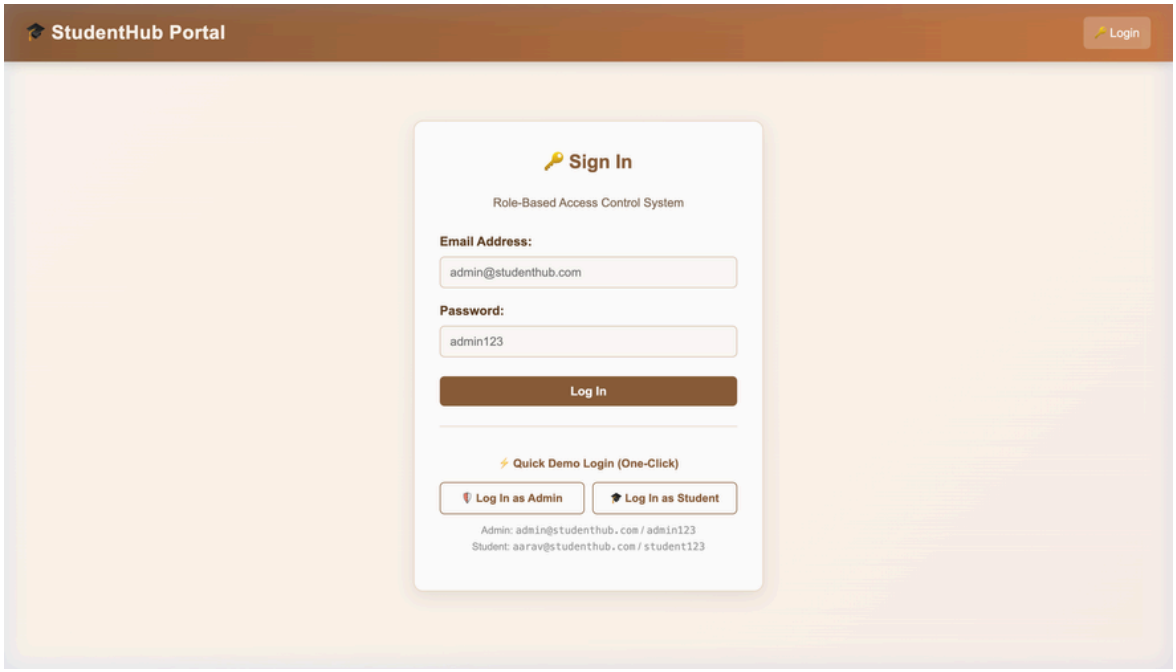
4. Are important admin actions recorded in audit_logs?

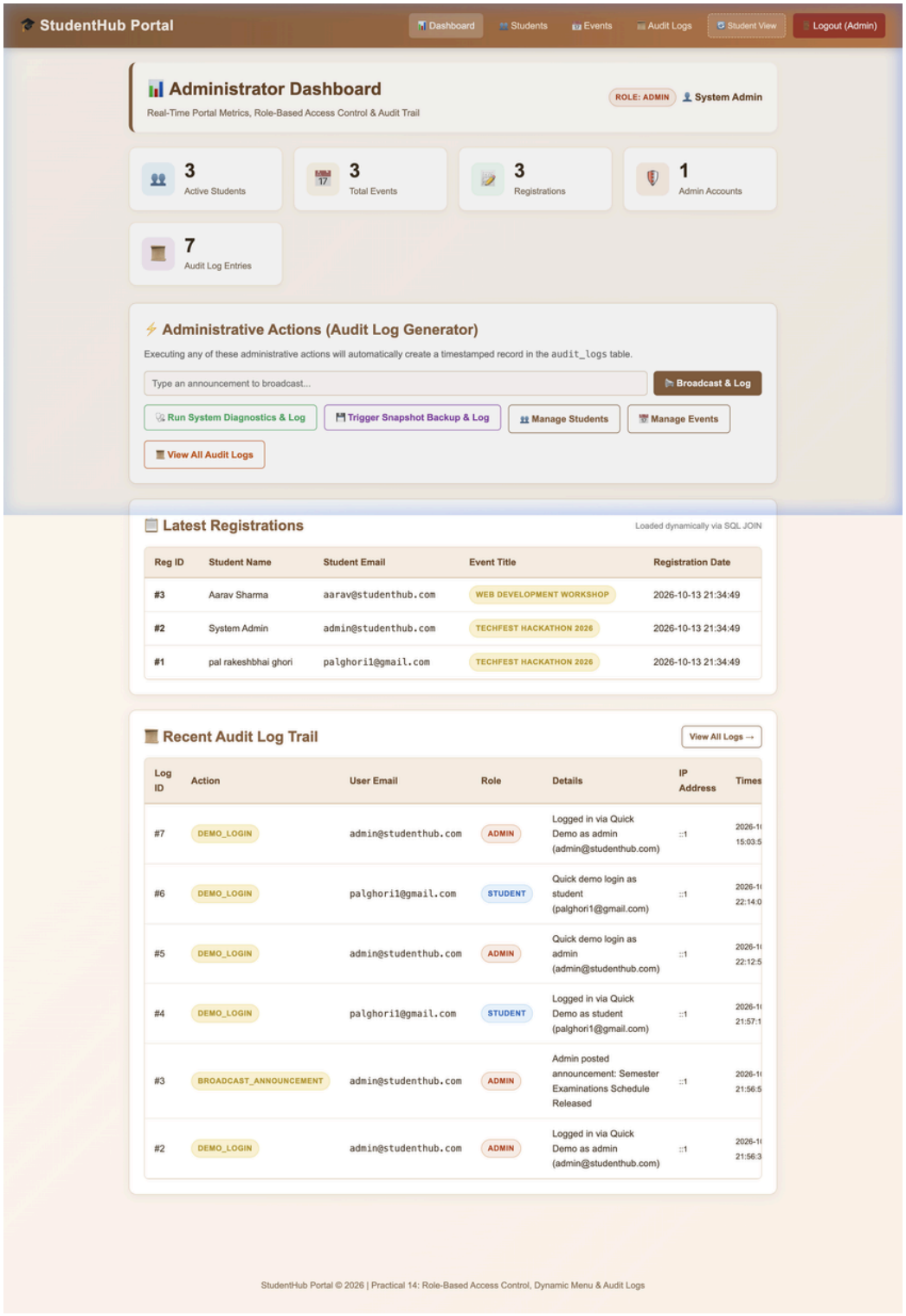
Yes. A dedicated audit_logs table records actions using record_audit_log($conn, $action, $details). Actions such as Admin Login, Logout, Announcement broadcasts, diagnostics, event addition/deletion, and role changes are stored with the user email, role, action type, description, client IP, and timestamp.

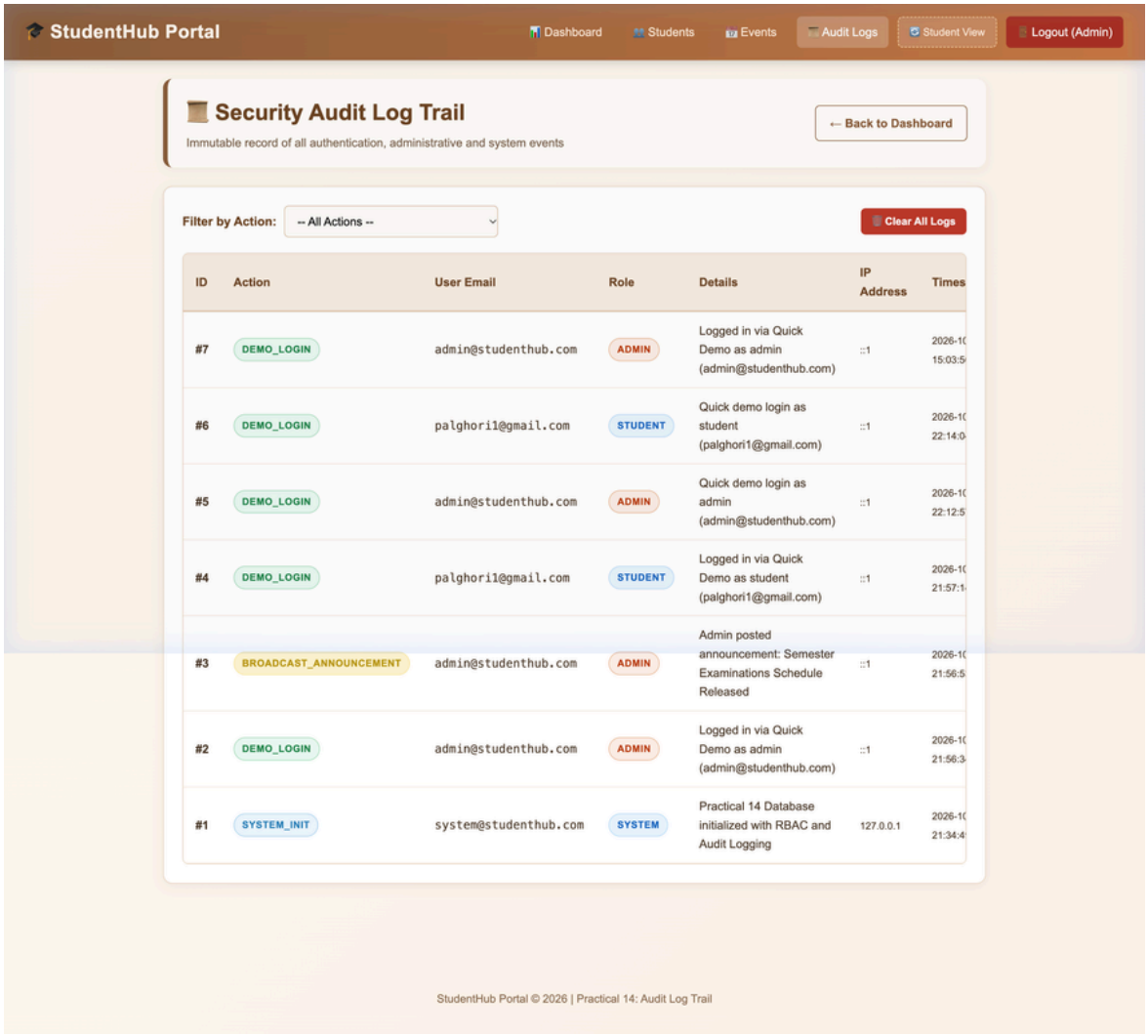
B. Implementation

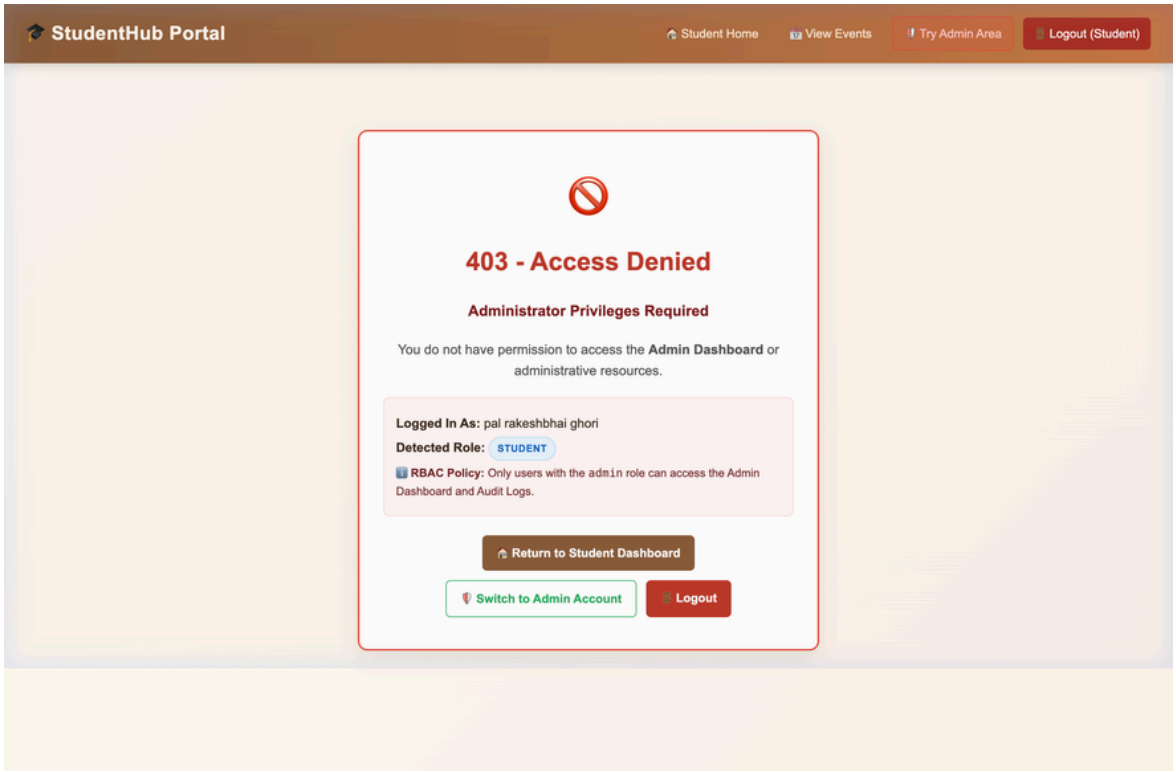
Procedural Architecture: Built with beginner-accessible procedural PHP (session_start(), mysqli_query(), mysqli_fetch_assoc()).
RBAC: Enforced server-side using $_SESSION['role'] checks on all protected routes.
Dynamic Menu: Reusable navbar.php component utilizing basic if/elseif conditions to render menu options according to role.
Audit System: Centralized record_audit_log() helper inserting structured event trails into the database.

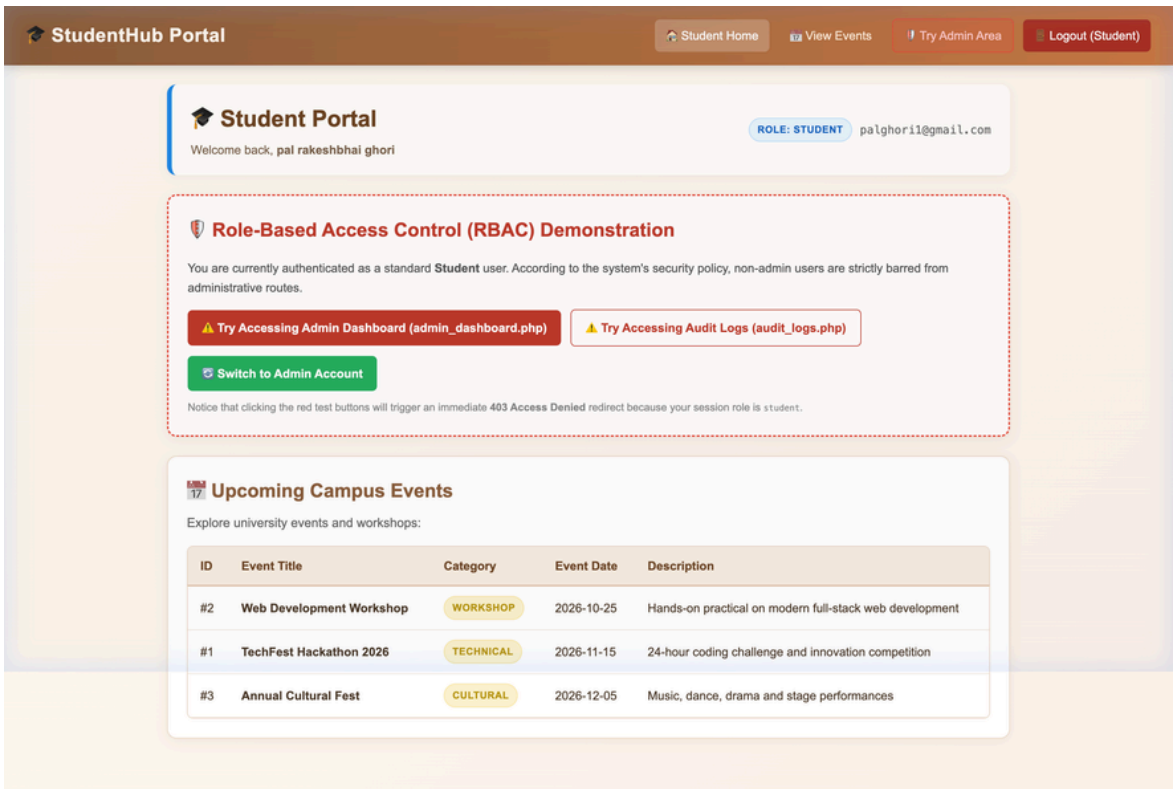
C. Evidence of Work (Screenshots)




StudentHub Portal
Dashboard
Students
Events
Audit Logs
Student View
Logout (Admin)
Administrator Dashboard
Real-Time Portal Metrics, Role-Based Access Control & Audit Trail
ROLE: ADMIN
System Admin
3
Active Students
3
Total Events
3
Registrations
1
Admin Accounts
7
Audit Log Entries
Administrative Actions (Audit Log Generator)
Executing any of these administrative actions will automatically create a timestamped record in the audit_logs table.
Type an announcement to broadcast...
Broadcast & Log
Run System Diagnostics & Log
Trigger Snapshot Backup & Log
Manage Students
Manage Events
View All Audit Logs
Latest Registrations
Loaded dynamically via SQL JOIN
Reg ID
Student Name
Student Email
Event Title
Registration Date
#3
Aarav Sharma
aarav@studenthub.com
WEB DEVELOPMENT WORKSHOP
2026-10-13 21:34:49
#2
System Admin
admin@studenthub.com
TECHFEST HACKATHON 2026
2026-10-13 21:34:49
#1
pal rakeshbhai ghori
palghori1@gmail.com
TECHFEST HACKATHON 2026
2026-10-13 21:34:49
Recent Audit Log Trail
View All Logs →
Log ID
Action
User Email
Role
Details
IP Address
Times
#7
DEMO_LOGIN
admin@studenthub.com
ADMIN
Logged in via Quick Demo as admin (admin@studenthub.com)
::1
#6
DEMO_LOGIN
palghori1@gmail.com
STUDENT
Quick demo login as student (palghori1@gmail.com)
::1
#5
DEMO_LOGIN
admin@studenthub.com
ADMIN
Quick demo login as admin (admin@studenthub.com)
::1
#4
DEMO_LOGIN
palghori1@gmail.com
STUDENT
Logged in via Quick Demo as student (palghori1@gmail.com)
::1
#3
BROADCAST_ANNOUNCEMENT
admin@studenthub.com
ADMIN
Admin posted announcement: Semester Examinations Schedule Released
::1
#2
DEMO_LOGIN
admin@studenthub.com
ADMIN
Logged in via Quick Demo as admin (admin@studenthub.com)
::1
StudentHub Portal © 2026 | Practical 14: Role-Based Access Control, Dynamic Menu & Audit Logs


StudentHub Portal
Dashboard
Students
Events
Audit Logs
Student View
Logout (Admin)
Security Audit Log Trail
Immutable record of all authentication, administrative and system events
← Back to Dashboard
Filter by Action:
-- All Actions --
Clear All Logs
ID
Action
User Email
Role
Details
IP Address
Times
#7
DEMO_LOGIN
admin@studenthub.com
ADMIN
Logged in via Quick Demo as admin (admin@studenthub.com)
::1
#6
DEMO_LOGIN
palghori1@gmail.com
STUDENT
Quick demo login as student (palghori1@gmail.com)
::1
#5
DEMO_LOGIN
admin@studenthub.com
ADMIN
Quick demo login as admin (admin@studenthub.com)
::1
#4
DEMO_LOGIN
palghori1@gmail.com
STUDENT
Logged in via Quick Demo as student (palghori1@gmail.com)
::1
#3
BROADCAST_ANNOUNCEMENT
admin@studenthub.com
ADMIN
Admin posted announcement: Semester Examinations Schedule Released
::1
#2
DEMO_LOGIN
admin@studenthub.com
ADMIN
Logged in via Quick Demo as admin (admin@studenthub.com)
::1
#1
SYSTEM_INIT
system@studenthub.com
SYSTEM
Practical 14 Database initialized with RBAC and Audit Logging
127.0.0.1
StudentHub Portal © 2026 | Practical 14: Audit Log Trail


StudentHub Portal
Student Home
View Events
Try Admin Area
Logout (Student)
403 - Access Denied
Administrator Privileges Required
You do not have permission to access the Admin Dashboard or administrative resources.
Logged In As: pal rakeshbhai ghori
Detected Role: STUDENT
RBAC Policy: Only users with the admin role can access the Admin Dashboard and Audit Logs.
Return to Student Dashboard
Switch to Admin Account
Logout


StudentHub Portal
Student Home
View Events
Try Admin Area
Logout (Student)
Student Portal
Welcome back, pal rakeshbhai ghori
ROLE: STUDENT
palghori1@gmail.com
Role-Based Access Control (RBAC) Demonstration
You are currently authenticated as a standard Student user. According to the system's security policy, non-admin users are strictly barred from administrative routes.
Try Accessing Admin Dashboard (admin_dashboard.php)
Try Accessing Audit Logs (audit_logs.php)
Switch to Admin Account
Notice that clicking the red test buttons will trigger an immediate 403 Access Denied redirect because your session role is student.
Upcoming Campus Events
Explore university events and workshops:
ID
Event Title
Category
Event Date
Description
#2
Web Development Workshop
WORKSHOP
2026-10-25
Hands-on practical on modern full-stack web development
#1
TechFest Hackathon 2026
TECHNICAL
2026-11-15
24-hour coding challenge and innovation competition
#3
Annual Cultural Fest
CULTURAL
2026-12-05
Music, dance, drama and stage performances

D. GitHub Repository Link:

https://github.com/palghori/STUDENT-HUB-PORTAL-WDF/tree/main/practical%214

E. Challenges Faced:

1. Session State Synchronization: Ensuring session destruction on logout completely resets both the PHP session array and browser cookie before redirecting.
2. Access Control Without Code Duplication: Creating a centralized access_denied.php handler and reusable navbar.php so role logic remains clean without repeating code across multiple pages.
3. Live Action Tracking: Ensuring audit logging occurs immediately alongside database operations (like INSERT/DELETE) without complicating the query flow.

F. Learning Outcomes:

1. Understood the fundamentals of Role-Based Access Control (RBAC) and server-side authorization beyond basic authentication.
2. Learned how to write SQL aggregate queries (COUNT(*)) and multi-table JOIN statements to feed real-time dashboards.
3. Learned how to implement Dynamic UI Navigation in PHP based on session variables.
4. Understood the importance of Audit Trails in web security for monitoring user activities and detecting unauthorized access attempts.

Faculty Remarks: